Paper No.	Paper/Patent Name	Link or PDF	Summary/Drawback
1.	Gas Leak Detector in Mining: Predictive Care	Becker Wholesale Mine Supply	methane (CH4), carbon monoxide (CO), hydrogen sulfide (H2S), and oxygen (O2) levels. If dangerous levels are detected, alarms ensures that data is not isolated but instead shared across the entire operation Does not contain health monitoring of a miner.
2.	Mine operation monitoring system	US9234426B2 - Mine operation monitoring system - Google Patents	Method of monitoring a mine operation comprises collecting map data associated with a mine operation. such a map tool has limited use in relation to mine operations since the photographic images quickly become too old.
3.	**Mining gas detection device suitable for remote detection**	Mining gas detection device suitable for remote detection - Eureka | Patsnap	This device is particularly well suited for deployment in underground mining operations where continuous, distributed gas monitoring is critical for personnel safety and early warning of hazardous leakages over extended distances from the central control room. **Limited Mobility**: The reliance on a fixed support leg, **Absence of Wireless Communication, Installation Complexity.**
4.	**Mining Gas Detectors: Gas Leak Detection Solutions for Miners**	Mining Gas Detectors: Gas Leak Detection Solutions for Miners | PK Safety	Detects methane (CH_4), carbon monoxide (CO), hydrogen sulfide (H_2S), oxygen-deficient environments, and even radon. Remote monitoring of difficult-to-access areas. Very costly 570$, connectivity issues underground. Maintenance is difficult. For large mines more devices are needed. No health monitoring.
5.	**Gas detection device for coal mine (PATENT)**	Gas detection device for coal mine - Eureka | Patsnap	The patent seems mainly about mechanical design (protection device, filtering, cleaning) rather than specifying *which* kinds of gas sensors are used. No health monitoring. fixed installation while the miners are mobile. Doesn't addresses the human interaction with the device.

6.	**IoT Based Smart Mining Helmet For Hazardous Gas Detection and Miner Health Monitoring**	IoT Based Smart Mining Helmet For Hazardous Gas Detection and Miner Health Monitoring	Alert for one miner only. No connectivity with the other miners. Alerting by buzzer only.
7.	Wearable health monitoring device	Wearable health monitoring device (Patent) | OSTI.GOV	wearable patient device. ECG monitoring,
8.	**Real-Time Monitoring of Underground Miners' Status Based on Mine IoT System**	Real-Time Monitoring of Underground Miners' Status Based on Mine IoT System - PMC	The device hovers a few millimeters above the skin, rather than touching it directly. This is especially useful for fragile skin (wounds, burns, ulcers. The device is geared toward measuring gases from or through the skin (skin flux, VOCs, CO_2, water vapor), not necessarily detecting external harmful gases. How is it alerting is not mentioned. No health monitoring. Cost and maintenance is an issue.
9.	**Smart Helmet for Air Quality and Hazardous Event Detection for the Mining**	Smart Helmet for Air Quality and Hazardous Event Detection for the Mining industry - M.Tech B.Tech Engineering Projects Thesis Research Help in New Delhi, INDIA	An LCD screen displays real-time data Employs IR sensors to detect helmet removal----seems unnecessary to me. Incorporates a DHT11 sensor to measure temperature and humidity. can affect the helmet's weight and comfort. No Data Transmission.
10.	**Coal Mine Safety And Health Monitoring System Using LoRa**	IARJSET-AITCON-6.pdf	These sensors transmit data wirelessly using LoRaWAN technology, system aims to alert users to dangerous mine environments. significant battery drain, increase the weight and bulk of the helmet, underground mining operations, wireless communication can be unreliable due to signal interference.
11.	A wireless sensor network for coal mine safety powered by modified localization algorithm	A wireless sensor network for coal mine safety powered by modified localization algorithm - ScienceDirect	WSN enables real-time data transmission, allowing for continuous monitoring and timely alerts. system is powered by a modified solar energy setup. affected by weather conditions and geographical location, leading to power shortages. wireless signals can be obstructed by rocks and other materials, leading to

			communication failures or delays in data transmission. NO health monitoring.
12.	**Intelligent Gas Detection Robot for Coal Mine Based on Image Recognition**	CN-114280270-A - Intelligent Gas Detection Robot for Coal Mine Based on Image Recognition | Unified Patents CN114280270A - 一种基于图像识别的智能煤矿用瓦斯气检测机器人 - Google Patents	gas detection robot. plate with two mounting boxes, each equipped with magnetic plates Multiple gas detection devices attached. Image Recognition Modules mounted on the outer walls. display panel to show real-time gas concentration levels. image recognition modules depends on lighting and environmental conditions. No health monitoring.
13.	A SMART HELMET FOR IMPROVING SAFETY IN MINING INDUSTRY		integrated system to provide supplemental oxygen in case of hazardous gas detection Sensors to detect impacts or collisions, alerting. Adding sensors and components can increase the weight and bulk of the helmet. We are detecting wider range of harmful gases. Health monitoring is not there.
14.	Iot Based Smart Helmet for Mining Industry Application	IJCRT2207508.pdf	**GPS Tracking , Panic Button.** ZigBee wireless communication is reliable up to about 100 meters. No health monitoring. Helmet integration may increase its weight and give discomfort.